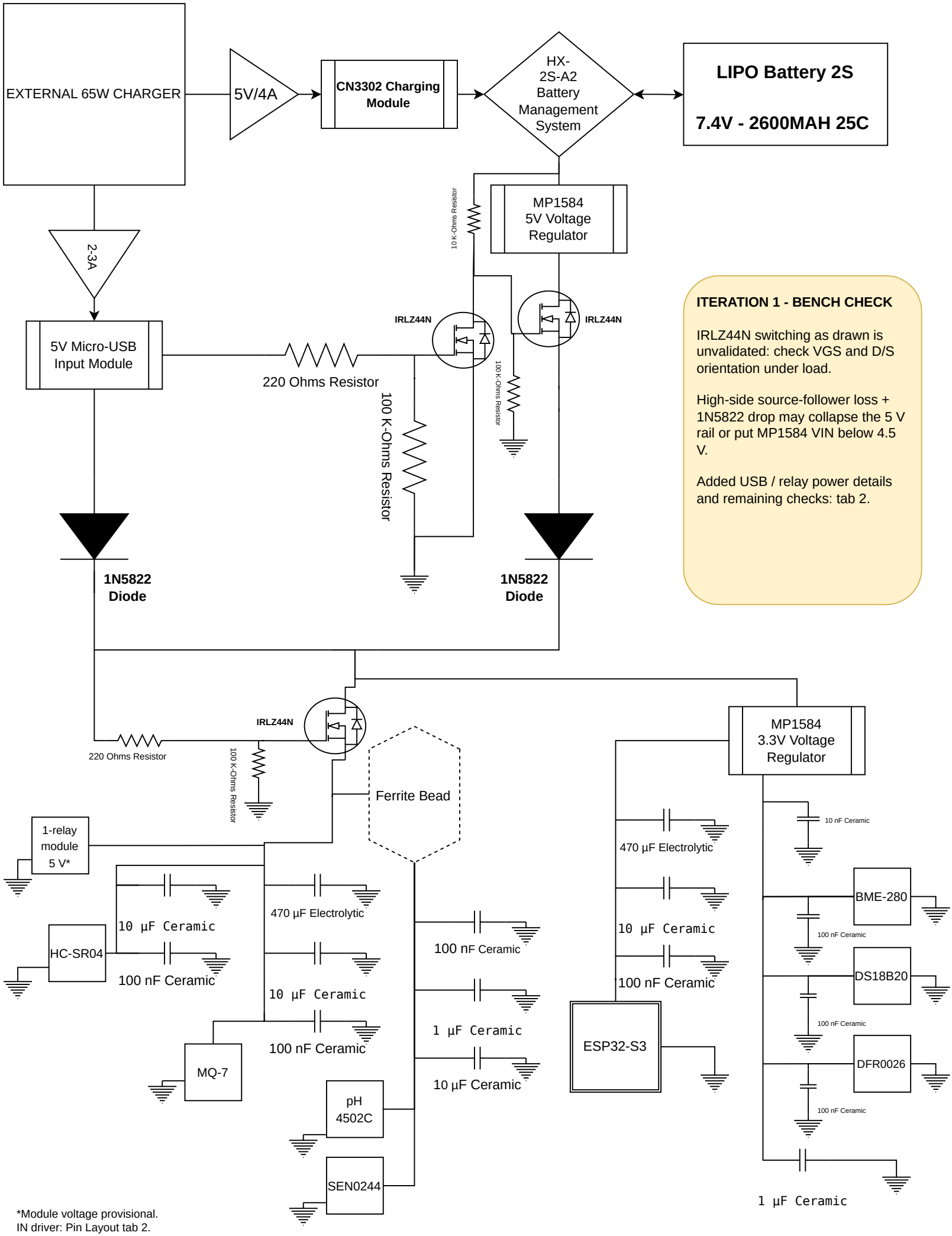


*Module voltage provisional.
IN driver: Pin Layout tab 2.

Power additions & first-iteration review

J1 programming Micro-USB
1 VBUS: NC / insulated
2 D- / 3 D+: Pin Layout tab 2
4 ID: NC
5 GND + shield: board GND

Existing board supply
3V3_SYS + GND
ESP32-S3, BOOT and EN pull-ups
EN RC and buttons: Pin Layout tab 2

Separate ports: J1 is for firmware/data. Keep its VBUS disconnected from the charger, battery and existing 5 V Micro-USB power input. Board must be powered separately for programming.

Existing 5 V load rail*
after source selection
5V_RELAY + GND

1-relay module, provisional 5 V
VCC / GND from existing rail
IN driven by NPN: Pin Layout tab 2

At relay VCC-GND: add 100 nF + 10 µF local decoupling.
Keep relay switching-current return away from analog-sensor / ADC return paths.
*Measure actual rail under relay load; its 5 V value is not yet validated.

1. Power switch headroom - unresolved
N-channel devices used high-side need gate voltage above the source for low resistance. Check the three IRLZ44N stages as drawn. Gate threshold is not full conduction; record gate, source, drain and load voltage on USB and battery power.

2. 5 V rail and 3.3 V startup
MP1584 specifies VIN >= 4.5 V. Diode and MOSFET losses can violate that limit. Measure switchover, relay steps and Wi-Fi bursts; confirm stable 3.3 V and at least 500 mA supply capability for the ESP32-S3.

3. Charging and connectors
Confirm the charger actually provides each specified 5 V output concurrently. Check connector / cable current ratings, CN3302 2S charge setting, BMS B+/BM/B-/P terminals and balancing capability. A 65 W label alone does not establish 5 V / 4 A availability.

4. Bare-chip PCB work still required
These are block / signal diagrams. Confirm exact ESP32-S3 package and add all supply-pin decoupling, exposed-pad GND, required flash, 40 MHz crystal, RF matching / antenna and strap configuration before PCB layout.

5. Sensor and relay qualification
Verify actual module output voltages and common GND. MQ-7 heater cycling, ADC input range, pH calibration, relay trigger polarity / flyback and contact-load suppression remain bench checks. See the accompanying review notes for details.